

Theory Assignment-1

Q1. Explain the HTTP Request–Response Cycle with a suitable diagram. Describe the major steps involved when a client (browser) requests a resource from a web server and receives a response.

Ans:

HTTP Request–Response Cycle

The **HTTP Request–Response Cycle** is the communication process between a client (browser) and a web server.

Major Steps:

1. **Client Request:** The user enters a URL in the browser.
2. **DNS Lookup:** The browser converts the domain name into the server's IP address.
3. **Connection:** The browser establishes a connection with the web server.
4. **HTTP Request:** The browser sends an HTTP request such as GET or POST to request a resource.
5. **Server Processing:** The server receives and processes the request and finds or generates the required resource.
6. **HTTP Response:** The server sends a response containing a status code, headers, and requested data.
7. **Browser Processing:** The browser receives and processes the response.
8. **Display:** The browser renders the content and displays the webpage to the user.

Q2. Briefly explain the purpose and key features of the following web technologies:

1. HTML
2. CSS
3. JavaScript

4. Bootstrap

5. jQuery

6. React.js

Ans:-

1. HTML (HyperText Markup Language)

- **Used to create the structure of web pages.**
- **Defines elements such as headings, paragraphs, images, links, tables, and forms.**
- **Uses tags like <html>, <head>, <body>, <h1>, and <p>.**

2. CSS (Cascading Style Sheets)

- **Used to style and design web pages.**
- **Controls colors, fonts, spacing, layouts, and responsiveness.**
- **Helps make websites attractive and user-friendly.**

3. JavaScript

- **Used to add interactivity and dynamic behavior to web pages.**
- **Can handle events, validate forms, modify HTML/CSS, and perform calculations.**
- **Runs mainly in the web browser.**

4. Bootstrap

- **A CSS framework used for designing responsive websites quickly.**
- **Provides ready-made buttons, forms, navigation bars, cards, and layouts.**
- **Supports responsive design for mobile, tablet, and desktop.**

5. jQuery

- **A JavaScript library that simplifies JavaScript programming.**
- **Makes DOM manipulation, event handling, animations, and AJAX easier.**
- **Uses shorter and simpler syntax than traditional JavaScript.**

6. React.js

- **A JavaScript library for building user interfaces.**

- **Developed and maintained by Meta.**
- **Uses reusable components and a virtual DOM.**
- **Commonly used for developing modern single-page applications.**

Q3. What are client-side technologies and server-side technologies? Explain the difference between them with suitable examples.

Ans:- Client-side technologies are technologies that run on the user's web browser. They are mainly used to create the appearance and interactivity of a website.

Examples: HTML, CSS, JavaScript, React.js.

Server-side technologies run on the web server. They handle business logic, database operations, authentication, and processing of requests.

Examples: Python, PHP, Java, Node.js, ASP.NET.

Difference

1. Client-side code runs in the browser, while server-side code runs on the server.
2. Client-side is mainly used for UI and user interaction, while server-side handles data processing and business logic.
3. Client-side technologies include HTML, CSS, and JavaScript, while server-side technologies include Python, PHP, Java, and Node.js.

Example: In a login system, JavaScript can validate the form on the client side, while the server checks the username and password against a database.

Q4. What are block-level elements in HTML? Explain their characteristics and provide at least three examples.

Ans:-

Block-Level Elements in HTML

Block-level elements are HTML elements that start on a new line and usually occupy the full available width of their parent element.

Characteristics:

- Start on a new line.
- Usually take up the full width available.
- Can contain other block-level and inline elements.

- Can have width, height, margin, and padding applied.

Examples:

1. <div> – Defines a general block/container.
2. <p> – Defines a paragraph.
3. <h1> to <h6> – Defines headings.
4. <section> – Defines a section of a webpage.

Q5. What are semantic elements in HTML? Explain their importance in web development and provide suitable examples.

Ans:-

Semantic Elements in HTML

Semantic elements are HTML elements that clearly describe the meaning and purpose of their content.

Importance:

- Improve code readability and structure.
- Help search engines understand webpage content.
- Improve accessibility for screen readers.
- Make websites easier to maintain and develop.

Examples:

- <header> – Defines the header of a page or section.
- <nav> – Defines navigation links.
- <main> – Defines the main content.
- <section> – Defines a section of content.
- <article> – Defines independent content.
- <footer> – Defines the footer.

Example:

```
<header>My Website</header>
```

```
<nav>Home | About | Contact</nav>
```

```
<main>
```

```
  <article>
```

```
<h1>Welcome</h1>
<p>This is my webpage.</p>
</article>
</main>
<footer>Copyright 2026</footer>
```

In short: Semantic elements make HTML meaningful, structured, accessible, and SEO-friendly.

Q6. Write HTML code to create the following table: Sr. No. Name Designation
Department 1 John Manager Sales 2 Smith Sr. Manager Marketing 3 Jane
Manager HR The table should include appropriate table headings, rows, and columns.

Ans:-

```
<!DOCTYPE html>
<html>
<head>
  <title>Employee Table</title>
</head>
<body>
<table border="1">
  <tr>
    <th>Sr. No.</th>
    <th>Name</th>
    <th>Designation</th>
    <th>Department</th>
  </tr>
  <tr>
    <td>1</td>
    <td>John</td>
    <td>Manager</td>
    <td>Sales</td>
```

```
</tr>
<tr>
  <td>2</td>
  <td>Smith</td>
  <td>Sr. Manager</td>
  <td>Marketing</td>
</tr>
<tr>
  <td>3</td>
  <td>Jane</td>
  <td>Manager</td>
  <td>HR</td>
</tr>
</table>
</body>
</html>
```

Q7. Write HTML code to create the following list using an appropriate combination of ordered/unordered lists:

- River
 - Ganga
 - Yamuna
 - Brahmaputra
 - Narmada

Ans:-

```
<!DOCTYPE html>
<html>
<body>
<ol>
  <li>River
    <ul>
```

```
<li>Ganga</li>
<li>Yamuna</li>
<li>Brahmaputra</li>
<li>Narmada</li>
</ul>
</li>
</ol>
</body>
</html>
```

Q8. Explain the different types of CSS used for applying styles to an HTML document. Describe each type with a suitable example.

Ans:-

Types of CSS

There are three types of CSS used to apply styles to an HTML document:

1. Inline CSS

CSS is written directly inside the HTML element using the style attribute.

Example:

```
<p style="color: blue;">Hello World</p>
```

2. Internal CSS

CSS is written inside the <style> tag in the <head> section of the HTML document.

Example:

```
<head>
  <style>
    p {
      color: blue;}
  </style>
</head>
```

3. External CSS

CSS is written in a separate .css file and linked to the HTML document using the <link> tag.

HTML:

```
<link rel="stylesheet" href="style.css">
```

style.css:

```
p {  
    color: blue;}
```

In short: Inline CSS is used for a single element, Internal CSS for one webpage, and External CSS for multiple webpages.

Q9. Differentiate between HTML, CSS, and JavaScript based on their role in developing a web page. Explain what each technology contributes to a website.

Ans:-

HTML, CSS, and JavaScript are the three main technologies used to develop web pages.

1. HTML (Structure)

HTML defines the structure and content of a webpage. It adds headings, paragraphs, images, links, forms, tables, etc.

Example: <h1>Welcome</h1>

2. CSS (Presentation)

CSS is used to style and design the webpage. It controls colors, fonts, spacing, layout, and responsiveness.

Example: h1 { color: blue; }

3. JavaScript (Behavior)

JavaScript adds interactivity and dynamic behavior to a webpage. It handles events, form validation, animations, and dynamic content.

Example: alert("Welcome!");

In short:

HTML → Structure

CSS → Style

JavaScript → Behavior/Interaction

Q10. Answer the following:

- What is the role of Bootstrap in web development?
- What is jQuery and why was it widely used in web development?

c. What is React.js and what problem does it solve in frontend development?

Ans:-

a. Role of Bootstrap in Web Development

Bootstrap is a front-end framework used to create responsive and attractive websites quickly. It provides ready-made CSS classes and components such as buttons, forms, navigation bars, cards, and grids.

b. jQuery

jQuery is a JavaScript library that simplifies tasks such as DOM manipulation, event handling, animations, and AJAX requests. It was widely used because it made JavaScript easier, shorter, and more compatible across different browsers.

c. React.js

React.js is a JavaScript library for building user interfaces. It uses reusable components and a Virtual DOM to efficiently update the webpage.

It solves the problem of managing complex and dynamic user interfaces by allowing developers to divide the UI into small, reusable components.